M|P|W

Physics A (H156, H556)

PHBx_Chapter 6 TA
Nick Papanikolaou

Please note that you may see slight differences between this paper and the original.

Candidates answer on the Question paper.

OCR supplied materials:
Additional resources may be supplied with this paper.

Other materials required:
- Pencil
- Ruler (cm/mm)

Duration: 45 mins

INSTRUCTIONS TO CANDIDATES

- Write your name, centre number and candidate number in the boxes above. Please write clearly and in capital letters.
- Use black ink. HB pencil may be used for graphs and diagrams only.
- Answer **all** the questions, unless your teacher tells you otherwise.
- Read each question carefully. Make sure you know what you have to do before starting your answer.
- Where space is provided below the question, please write your answer there.
- You may use additional paper, or a specific Answer sheet if one is provided, but you must clearly show your candidate number, centre number and question number(s).

INFORMATION FOR CANDIDATES

- The quality of written communication is assessed in questions marked with either a pencil or an asterisk. In History and Geography a *Quality of extended response* question is marked with an asterisk, while a pencil is used for questions in which *Spelling, punctuation and grammar and the use of specialist terminology* is assessed.
- The number of marks is given in brackets **[ ]** at the end of each question or part question.
- The total number of marks for this paper is **37**.
- The total number of marks may take into account some 'either/or' question choices.

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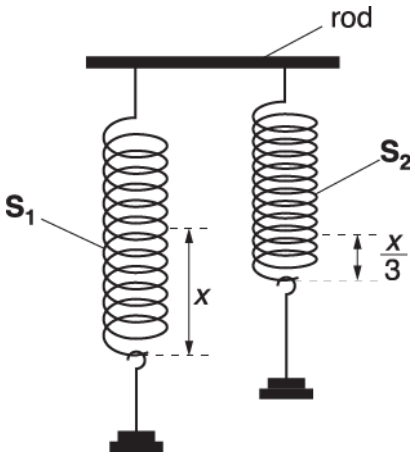
1 The top ends of two springs, S_1 and S_2, are attached to a rod.



A mass is hung from the bottom end of S_1. The extension of S_1 is x. The elastic potential energy in the spring is E. The same mass is hung from the bottom end of S_2. The extension of S_2 is $\frac{x}{3}$.

What is the elastic potential energy in the spring S_2?

A $\frac{E}{9}$

B $\frac{E}{3}$

C $3E$

D $9E$

Your answer ☐

[1]

2 Two guitar strings of equal length, but of different thickness, are under the same tension. The strings are made of the same material.

The thinner string has a diameter of 0.20 mm and the thicker string has a diameter of 0.80 mm.

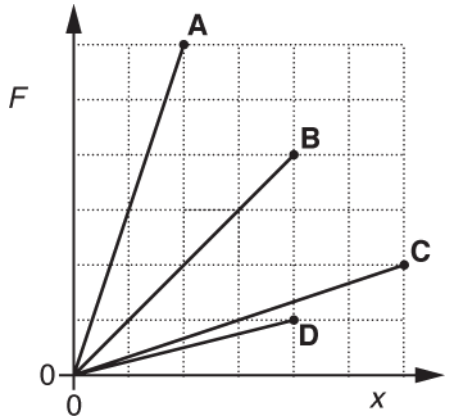
What is the value of the ratio $\frac{\text{elastic potential energy in the thinner string}}{\text{elastic potential energy in the thicker string}}$?

A 0.125
B 0.25
C 4.0
D 16

Your answer ☐

[1]

3 The force *F* against extension *x* graphs for four different wires **A**, **B**, **C** and **D** up to their breaking points are shown below.



Which wire has the greatest work done on it before it breaks?

Your answer ☐ **[1]**

4 One end of a wire is fixed to the ceiling and a 3.0 kg object is suspended from its other end. The wire has diameter 0.62 mm and negligible mass.

What is the tensile stress in the wire?

A 1.5×10^4 Pa

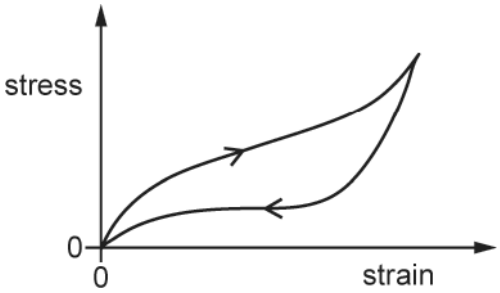
B 2.5×10^6 Pa

C 2.4×10^7 Pa

D 9.7×10^7 Pa

Your answer [1]

5 The stress against strain graph for a particular material is shown below.



Which term is correct for a property of this material?

A brittle
B ductile
C elastic
D plastic

Your answer [1]

6 A student investigated four different springs **A, B, C** and **D**.
A mass of different weight is hung from each spring and the extensions are measured.

The results are shown in the table below.

	Weight hung from spring / N	Extension of spring / cm
A	2.0	5.2
B	4.0	4.6
C	6.0	3.5
D	8.0	1.5

All of the springs obey Hooke's law.

Which spring has the **largest** elastic potential energy?

Your answer ☐ **[1]**

7 A student is recording results in a table.
The calculated value for the stress in a wire is 7 260 000 Pa.

Which of the following is correct for the heading in the student's table and the value of the stress written to **2** significant figures?

	Heading in table	Value of stress
A	stress / Pa	7 200 000
B	stress / kPa	730
C	stress / MPa	7.2
D	stress/10^6 Pa	7.3

Your answer ☐ **[1]**

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8 A group of students are investigating the loading and unloading of glass and rubber. Glass is a brittle material and rubber is a polymeric material.

Sketch the stress against strain graphs for the loading and unloading of glass and rubber on **Fig. 6**.

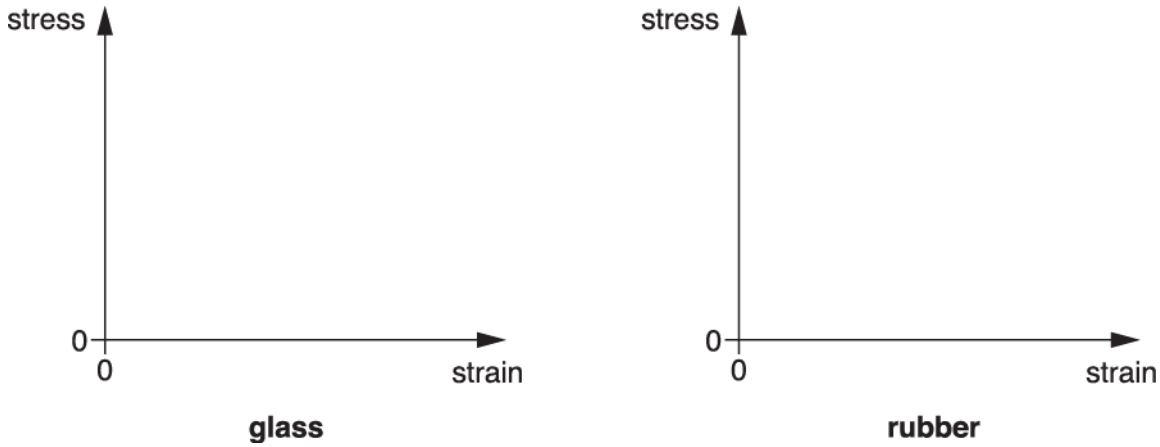

Fig. 6

[2]

9 The stress in a concrete pillar is 1.1×10^5 Pa. The original length of the pillar was 2.3 m. The Young modulus of concrete is 1.4×10^{10} Pa.

Calculate the compression x of the pillar.

x = _____ m **[3]**

10(a) The extension of a metal wire is x when the tension in the wire is F. The table in **Fig. 23.1** shows the results from an experiment, including the stress and the strain values.

F / N	x / 10^{-3} m	stress / 10^{7} Pa	strain / 10^{-3}
1.9	0.4	1.73	0.20
4.0	0.8	3.50	0.40
5.9	1.2	5.21	0.60
8.0		7.00	0.80
9.0	1.8	7.95	0.90

Fig. 23.1

Complete the table by determining the extension when the tension is 8.0 N.

[1]

(b) **Fig. 23.2** shows a graph of stress against strain for the metal.

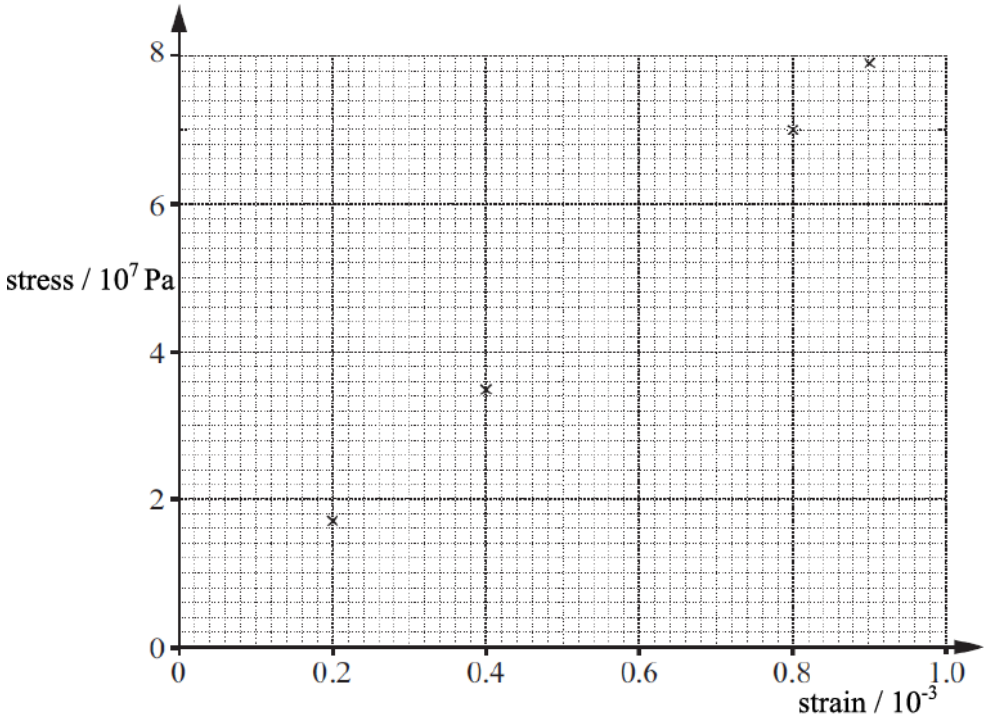

Fig. 23.2

(i) On **Fig. 23.2,** plot the data point corresponding to the tension of 5.9 N and draw the line of best fit through all the data points.

[1]

(ii) Use **Fig. 23.2** to determine the Young modulus of the metal.

.

Young modulus = _____ Pa **[2]**

(c) The micrometer screw gauge used to determine the diameter of the wire had a zero error. The diameter recorded by a student was larger than it should have been.

Discuss how the actual value of the Young modulus would differ from the value calculated in **(b)(ii)**.

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[3]

11 An engineer is investigating the tension in a steel cable supporting a uniform wooden plank as shown in Fig. 4.

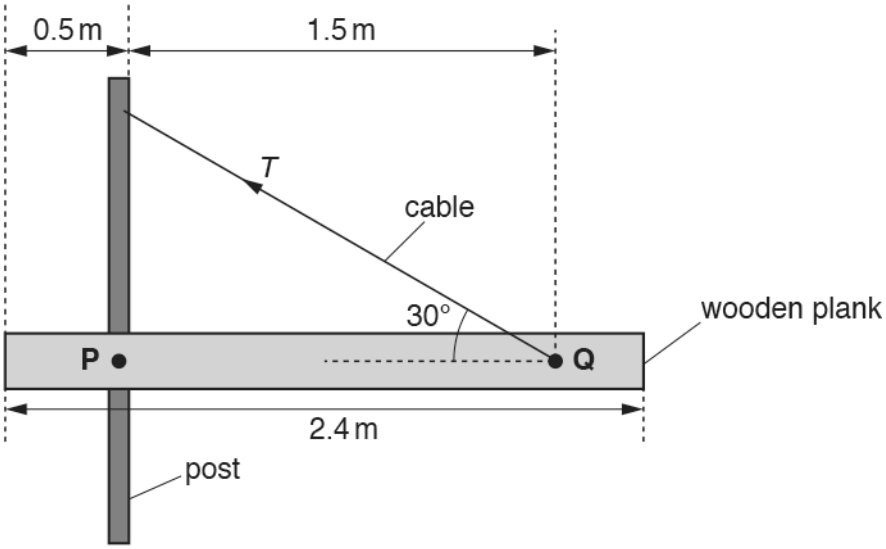

Fig. 4 (not to scale)

The plank is 2.4 m long and has a mass of 50 kg. It is pivoted at point **P** to a vertical post. The cable is fixed to the plank at point **Q** and to the vertical post as shown in Fig. 4. The cable is at an angle of 30° to the plank. The plank is in equilibrium and resting in a horizontal position.

The original length of the steel cable is 1.73 m and it has a cross-sectional area of 11.0 mm^2.
The Young modulus of steel is 210 GPa.
Calculate the extension *x* of the cable shown in Fig. 4.

x = ... m **[3]**

12(a) A metal wire has length 2.2m and cross-sectional area of $1.4 \times 10^{-7} m^2$. One end of the wire is fixed to the ceiling and a load of weight 49N is attached to the other end so that the wire is vertical.
The Young modulus of the metal is 180GPa.

The wire obeys Hooke's law.
Calculate

(i) the stress σ in the wire

σ = ... Pa **[2]**

(ii) the strain ε of the wire

ε = ... [2]

(iii) the extension x of the wire

x = ... m [2]

(iv) the elastic potential energy E of the wire.

E = ... J [2]

(b) The wire in **(b)** is replaced by a wire of the same metal and length but double the diameter. The same load is attached to the wire.

State and explain the change, if any, to the elastic potential energy of the wire.

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[2]

13 A crane raises a mass of 3000 kg through a height of 12 m in 40 seconds with an efficiency of 60%.

The crane cable is made of an alloy. The stress-strain curve for the cable is shown in Fig. 25.

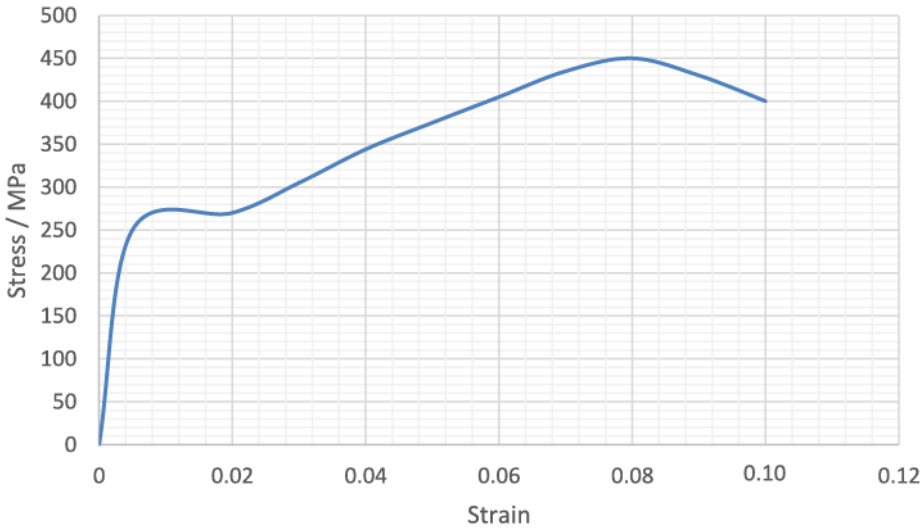

Fig. 25

(i) Use the graph to estimate the stress in the cable at its elastic limit.

stress = _____ MPa **[1]**

(ii) A cable of diameter 0.090 m has a maximum working load allowance of 1.1×10^6 N.

By calculating the maximum tensile stress allowed, suggest and explain why the working load allowance is 1.1×10^6 N.

[4]

END OF QUESTION PAPER

Question			Answer/Indicative content	Marks	Guidance
1			B	1	Examiner's Comments All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions.
			Total	1	
2			D	1	Examiner's Comments All of the questions showed a positive discrimination, and the less able candidates could access the easier questions. The questions in Section A do require careful reading and scrutiny. Candidates are advised to reflect carefully before recording their response in the box. Candidates must endeavour to use a variety of quick techniques when answering multiple choice questions.
			Total	1	
3			B	1	
			Total	1	
4			D	1	
			Total	1	
5			C	1	
			Total	1	
6			C	1	
			Total	1	
7			D	1	
			Total	1	
8			Glass: A straight line from the origin.	B1	Ignore arrows

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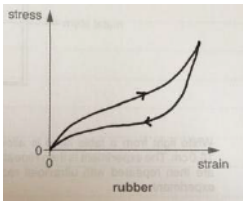
Question			Answer/Indicative content	Marks	Guidance
			Rubber: A correct sketch for loading and unloading sections, with the graph starting and finishing at the origin.  	B1	Allow either arrows or labelled curves Examiner's Comments Candidates needed to indicate the loading and unloading directions on the rubber curve. For the rubber curve, a number of candidates indicated that the rubber was increasing its strain when the load was being removed. Where a straight line is intended, a ruler should be used.
			Total	2	

Question			Answer/Indicative content	Marks	Guidance
9			$1.4\times10^{10} = \frac{1.1\times10^{5}}{\text{strain}}$ (Any subject) / strain = 7.86×10^{-6} $x = \frac{1.1\times10^{5}}{1.4\times10^{10}}\times2.3$ $x = 1.8 \times 10^{-5}$ (m) or $1.1 \times 10^{5} = \frac{6500}{A}$ / A = 0.059 (m^2) $(F = \frac{EAx}{L})$; $6500 = \frac{1.4\times10^{10}\times0.059\times x}{2.3}$ (Any subject) $x = 1.8 \times 10^{-5}$ (m)	C1 C1 A1 C1 C1 A1	Possible ECF from (a) **Examiner's Comments** This was a notable success for most of the candidates. Examiners were pleased to see a range of techniques being used to get to the correct answer of 1.8×10^{-5} m. Most of the answers were well-structured and showed good use of calculators. A small number of candidates correctly calculated the strain but then struggled to rearrange the expression for strain to determine the compression.
			Total	3	

Question			Answer/Indicative content	Marks	Guidance
10	a		1.6 ($\times 10^{-3}$ m).	B1	
	b	i	Data point plotted to within ± ½ small square and correct line of best fit though all the data points.	B1	
		ii	Gradient of line determined.	M1	
		ii	E = gradient = (8.8 ± 0.1) × 10^{10} (Pa).	A1	Allow 1 mark for (8.8 ± 0.1) × 10^{n} Pa; where $n \neq 10$
	c		The actual cross-sectional area will be smaller.	B1	
			The actual stress values on the graph will be larger (because stress ∝ area^{-1})	B1	
			The gradient of the graph will be larger; hence the Young modulus of the metal must be larger than the student's value.	B1	
			Total	7	

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Question			Answer/Indicative content	Marks	Guidance
11			$x = \frac{TL}{EA}$	C1	Note *x* must be the subject
			$x = \frac{460 \times 1.73}{210 \times 10^9 \times 11 \times 10^{-6}}$	C1	Allow alternative methods e.g. determines stress (4.18×10^7 Pa) C1 determines strain (1.99×10^{-4}) C1 determines *x*
			$x = 3.45 \times 10^{-4}$ (m)	A1	Allow 3.4, 3.5, 3.43, 3.44 Allow 2 marks for 3.45×10^n <u>Examiner's Comments</u> This question required candidates to carry out several calculations. Good candidates would start by combining the definitions of stress and strain with the definition of Young modulus to give $x = \frac{TL}{EA}$ A significant number of candidates made a power of ten (POT) error either with 210 GPa or with the area of 11.0 mm^2. Many candidates wrote the latter as 11×10^{-3} m^2. Other lower ability candidates tried calculating the area from this value. Some candidates correctly determined the stress, then the strain and then the extension.
			Total	3	

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Question			Answer/Indicative content	Marks	Guidance
12	a	i	$\frac{49}{1.4 \times 10^{-7}}$ 3.5×10^{8} Pa	C1 A1	
		ii	$\frac{3.5 \times 10^{8}}{180 \times 10^{9}}$ 0.0019	C1 A1	**Allow ECF** from **(b)(i)** **Ignore** units
		iii	0.0019 × 2.2 0.0042 (m)	C1 A1	**Allow ECF** from **(b)(i)** and **(ii)** **Allow** 0.0043 (m)
		iv	$\frac{1}{2} \times 49 \times 0.0043$ 0.10 (J)	C1 A1	**Allow ECF** from **(b)(i)**, (ii) and **(iii)** **Do not accept** 1 sf
	b		Area increases by a factor of four / extension decrease by factor of four Elastic strain energy decreases by a factor of four	M1 A1	
			Total	**10**	
13		i	(230 ± 40) M Pa	B1	
		ii	Stress = 1.1×10^{6} / (Π × 0.045^{2}) = 173 MPa	C1	
		ii	173 MPa < 230 MPa	A1	AW (ecf)
		ii	So will not stretch too much in use	B1	Allow any sensible contextual suggestion
		ii	Less chance of permanent deformation or fatigue	B1	
			Total	**5**	

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